

Sequences and Series

ANSWER KEY



Sequence limits

1 Determine whether each sequence converges, and find the limit if it does:

a $a_n = \frac{3n+1}{n+2}$ Converges to 3 (divide by n).

b $a_n = \frac{n^2}{2^n}$ Converges to 0 (exponential growth dominates polynomial).

c $a_n = (-1)^n$ Diverges (oscillates between -1 and 1).

2 Evaluate $\lim_{n \rightarrow \infty} (1 + \frac{3}{n})^n$.

This is the standard limit form giving e^3 .

Series and partial sums

3 Determine whether $\sum_{n=1}^{\infty} \frac{1}{n}$ converges or diverges, and name the test or fact that justifies your answer.
Diverges — this is the harmonic series, a standard example of a divergent series with terms going to 0.

4 Evaluate the geometric series $\sum_{n=0}^{\infty} 3(\frac{2}{5})^n$.
 $|r| = \frac{2}{5} < 1$: $S = \frac{3}{1-2/5} = 5$.

5 Use partial fractions and telescoping to evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.
 $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$. The partial sums telescope: $S_N = 1 - \frac{1}{N+1} \rightarrow 1$.